Adaptive Effective Receptive Field Convolution for Semantic Segmentation of VHR Remote Sensing Images

Xi Chen, *Member, IEEE*, Zhiqiang Li, Jie Jiang, Zhen Han, Shiyi Deng, Zhihong Li, Tao Fang, Hong Huo, *Member, IEEE*, Qingli Li, *Senior Member, IEEE*, and Min Liu

***Abstract*— Convolutional neural networks (CNNs) have facilitated impressive improvements in the semantic segmentation of very high-resolution (VHR) remote sensing images. The success of semantic segmentation depends on an effective receptive field (RF) large enough to cover the entire object. Popular methods to enlarge the effective RF include dilated filters, subsampling operations, and stacking layers. Unfortunately, the methods are inefficient or able to cause grid artifacts. Moreover, although the object sizes vary greatly in remote sensing images, the size of the RF cannot reach a compromise between small and large objects. To tackle these problems, we propose adaptive effective receptive convolution (AERFC) for VHR remote sensing images. AERFC adaptively controls the sampling location of convolution and automatically adjusts the effective RF without significantly increasing the parameter number and computational cost. Thus, AERFC reduces the training difficulty, decreases overfitting risk, and reserves details in VHR images. AERFC is also integrated with spatial pyramid pooling (SPP) to aggregate diverse multiscale features for exploring contextual information. Experimental results of the quantitative and qualitative evaluation over four benchmark data sets show that AERFC outperforms state-of-the-art methods.**

***Index Terms*— Field of view, filter, kernel, semantic contextual information.**

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Xi Chen is with the Key Laboratory of Geographic Information Science (Ministry of Education), School of Geographic Sciences, Shanghai 200241, China, and also with the Shanghai Key Laboratory of Multidimensional Information Processing, East China Normal University, Shanghai 200062, China.

Zhiqiang Li, Jie Jiang, Zhihong Li, and Min Liu are with the Key Laboratory of Geographic Information Science (Ministry of Education), School of Geographic Sciences, East China Normal University, Shanghai 200241, China (e-mail: lizhiqiangmy198267@163.com).

Zhen Han is with Software Engineering Institute, East China Normal University, Shanghai 200062, China.

Shiyi Deng is with Electronic and Information Engineering Institute, Hainan University, Hainan 570228, China.

Tao Fang and Hong Huo are with the Automation Department, Shanghai Jiao Tong University, Shanghai 200240, China.

Qingli Li is with the Shanghai Key Laboratory of Multidimensional Information Processing, East China Normal University, Shanghai 200062, China.

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I. Introduction

SEMANTIC segmentation formulates a pixelwise prediction for remote sensing images, which can be used in thematic mapping [1], vehicle extraction [2], high-definition (HD) maps for autonomous driving [3], and so on. Consequently, semantic segmentation is among fundamental topics in the remote sensing field, especially for very high-resolution (VHR) remote sensing images [4]. VHR images have fine-grained spatial details with only a few spectral channels. Fine-grained details lead to a relatively small spectral between-class variability and a relatively large within-class variability. On the other hand, only a few spectral channels result in less distinctive spectral signatures [4]. The two problems are serious threats to the semantic segmentation of VHR images.

To improve performance, semantic segmentation requires accurate feature representation, where the main information of VHR images is represented as shape, texture, and context, which are described by many handcrafted features [5]. Usually, it is hard to know in advance how the features can be combined or mapped for optimal semantic segmentation [6]. In addition, the feature extractors often require hyperparameters, such as gray-level co-occurrence matrices and sparse coding. These tuned features usually fail to generalize well on various data sets because the features cannot accurately describe the hidden semantic information. Consequently, they cannot meet the requirements of real-life applications [7]. These features should be replaced with features that can automatically be tuned for the target data sets [8].

Recently, useful geographical features have been widely automatically extracted by convolutional neural networks (CNNs) [9]. CNNs discard heuristic feature design and replace traditional filter bank approaches with a hierarchical architecture. Specifically, they implicitly capture features with filters by sharing weights and local connectivity to discover patterns at all locations in the VHR images [4]. The first convolution layer of CNNs learns simple features describing edges, similar to Gabor texture filters. Early layers extract small-scale features, including edges, shapes, and stripes from nearby pixels. The following layers learn more complex

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features by nonlinearly combining the previous features. The complex features are high-level abstract feature representations with various complexity and scales. With an increase in the receptive field (RF) (field-of-view) of the neurons, the complexity and level of the abstraction of the features continue to increase, and CNNs start to detect global structures/shapes over simple texture patterns [10].

A large RF is very ubiquitous in CNNs, such as in GoogleNet [11] or ResNet [12], because a large RF can significantly improve the prediction [13]. Generally, the pixels in the RF of CNNs provide unequal contributions [10], where a small region in the RF gathers the majority of the effective information [14]. The values in the effective area of the RF, i.e., the effective RF, can be calculated as the gradient of the response with respect to intensity perturbations [10], [15]. The theoretical RF is the RF, which can theoretically take effect. The effective RF is smaller than the theoretical RF. The limited effective RF leads the final feature map to obtain little context information, thus resulting in more false positives [16]. For better performance, the effective RF should be large enough to cover the entire object [17].

Generally, the effective RF size increases linearly with the square root of the RF size [18]. This increase is inefficient. A more straightforward and efficient way is to propose a new convolution. The typical example is a dilated filter, whose effective RF is increased by inserting holes between each pixel in the convolutional filter [19]. The convolution can be used to extract the multiscale context features [20]. For performance improvement, a dilated filter should benefit the extraction of all kinds of features, including the shape and texture [21], [22]. However, the dilated filter brings about gridding artifacts [23], which hampers the performance.

In addition, there are three challenges for popular convolutions to segment very small objects in large-scale remote sensing images, similar to object detection [16]. First, the ultralarge size of the effective RF makes the existing neural networks too slow for practical use. Second, the clustered and complex object distribution increases false positive rates. Finally, extremely small objects increase the difficulty of accurate segmentation. Therefore, it is necessary to automatically determine the RF size.

Unfortunately, to the best of our knowledge, there has been little work on new convolutions or filters to enlarge the effective RF for the semantic segmentation of VHR images. In this article, we propose adaptive effective RF convolution (AERFC) to extract more accurate features. Several key spatial sampling locations in the dilated filter are filled with trainable parameters to fit the potential edges of objects. The edges help to make up the typical shape of objects. In addition, AERFC can capture the regular arrangement of texture. Therefore, the kaleidoscope filter can approach the large-size regular convolution filter, which not only has a large RF but also has better capability to extract the context, shape, and texture features compared to the dilated filter.

The main contributions are listed as follows.

1) This is the first effort to address the effective RF for semantic segmentation of VHR remote sensing images. The effective RF convolutions should extend the RF without resolution loss.
2) A novel convolution, AERFC, is proposed to enlarge the effective RF and alleviate the gridding artifacts arising from the dilated filter. AERFC automatically captures the information in the original nonzero filter taps of the dilated filter and determines the effective RF for small or large objects without significantly increasing the number of extra parameters.
3) The convolutional network replacing dilated filters with AERFC surpasses the state-of-art methods over the ISPRS Vaihingen and ISPRS Potsdam data sets. To further demonstrate the effectiveness of our method, we also conduct experiments over two benchmark data sets, PASCAL VOC 2012 and Cityscapes. The results also verify the superiority of our method.

The remainder of our article is organized as follows. Section II presents a brief introduction to the related work. Section III provides the proposed method. The experimental results and discussion are shown in Section IV. Finally, Section V concludes our article.

II. Related Work

This section reviews the related methods that have been widely used, not limited to the remote sensing community. The existing methods can be divided into three categories: RF, dilated filter, and deformable convolution.

A. Receptive Field

The size of the effective RF is crucial because the output should respond to large enough regions in images to capture the information of large objects [10]. An effective way to increase the effective RF is the employment of downsampling in the contracting path [13]. Group pointwise and depthwise dilated separable convolutions can also be used to learn representations from a large effective RF [24]. The effective RF of the Zeiler and Fergus net is 171 pixels, which is suitable for ship detection [25]. Increasing the number of convolutional layers with downsampling can increase the size of the effective RF in the convolutional layers to learn enough contextual information [26]. By downsampling in the contracting path of U-net, the effective RF increases significantly for inverse scattering problems to improve the prediction [13]. An RF block module can be assembled to the top of a neural network for detection [27]. Trident Network generates scale-specific feature maps based on a parallel multibranch architecture, where each branch shares the same parameters with different RFs [28]. Remote sensing region-based CNN can inhibit false positives of ultrasmall objects. This network comprises a backbone with lightweight residual structure, an intermediate global attention block, a final classifier, and a detector [16]. To denoise the hyperspectral images, a 3-D atrous denoising CNN is proposed to extract feature maps along spatial and spectral dimensions with multibranch and multiscale structures [29]. To tackle sparse inputs, such as sparse laser scan data, a sparsity invariant CNN [30] is proposed to evaluate inputs by a sparse convolution that explicitly considers the

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location of invalid values. Compared with regular convolution, it has more effective RFs for sparse data.

B. Dilated Filter and Atrous Spatial Pyramid Pooling

The dilated filter is also called atrous convolution. The dilated filter is applied with rates determined by the desired output stride value, where the atrous rate refers to the stride [31]. The filter's rate is set to be larger than 1, where the weights are kept at sparse sampling locations. This convolution equals upsampled filters by inserting holes into the filters or sparsely sampling feature maps. Dilated filter generalizes the Kronecker-factored convolutional filters and exponentially enlarges the RF sizes without resolution reduction in the resolution of the feature maps. In this way, the dilated filter explicitly controls the density of feature responses. The dilated filter effectively incorporates a larger context by changing the atrous rates with the remaining computation complexity, without increasing the parameter number and the computation costs, compared to a regular convolution with larger filters [20], [31], [32]. A series of dilated filters can be combined to recover full resolution feature maps more densely to aggregate multiscale context instead of using deconvolutional layers [20]. Three parallel dilated filter operations are utilized in DeepLabv3 with various rates on top of the feature maps, which consist of atrous spatial pyramid pooling (SPP) (ASPP), to further describe the context at multiple scales. Similar to SPP [33], ASPP resamples features at various scales to accurately and efficiently classify regions. ASPP with various atrous rates can capture multiscale context information. As the sampling rate increases, the number of weights applied to the valid feature region decreases.

C. Deformable Convolution

Deformable convolution is a generalization of the dilated filter. Deformable convolution augments the spatial sampling locations with additional offsets, which are learned for the target tasks without additional supervision. Deformable convolution can be embedded in existing CNNs and trained end-to-end by a back-propagation procedure [15], [18]. A deformable region-based fully convolutional network (FCN) (R-FCN) adds offsets to the convolution layers and applies feature mapping to unfixed locations with a pretrained R-FCN by developing an aspect ratio-constrained nonmaximum suppression for a remedy of the increase in false lines [34]. Multibranch CNN employs a series of deformable CNNs for representative feature extraction and fuses multitemporal and multisensor Sentinel images [35].

III. Methodology

In this work, the AERFC convolutional network (AERFC ConvNet) is proposed for the semantic segmentation of VHR images. In this section, we will describe four important aspects of AERFC ConvNet, including: 1) AERFC; 2) AERFC-SPP; 3) modified Xception with AERFC; and 4) AERFC ConvNet.

A. Adaptive Effective Receptive Field Convolution

Let $\mathbf{x}$ be the input feature map, $\mathbf{w}$ be the weight map of a filter $\mathcal{R}$, $\mathbf{y}$ be the output feature map of convolution, and p_0 be every location in $\mathbf{y}(p_0)$. A convolution can be written as

$$\mathbf{y}(p_0) = \sum_{p_i \in \mathcal{R}} \mathbf{w}(p_i) \cdot \mathbf{x}(p_0 + p_i) \tag{1}$$

where p_i enumerates all the locations of $\mathbf{w}$ in $\mathcal{R}$. The locations p_i', p_i'', and p_i''' are generated by mirroring p_i. As shown in Fig. 1, AERFC consists of two steps: 1) assigning 16 sampling weights $\mathbf{w}(p_i)$, $\mathbf{w}(p_i')$, $\mathbf{w}(p_i'')$, $\mathbf{w}(p_i''')$ on p_i, p_i', p_i'', and p_i''', where $i = 1, 2, 3, 4$, into a 3×3 dilated filter $\mathcal{R}_{\text{Dilated}(3\times3)}$ to generate AERFC filter $\mathcal{R}_{\text{AERFC}}$; 2) summation of feature values of $\mathbf{x}$ weighted by $\mathbf{w}$ of $\mathcal{R}_{\text{AERFC}}$.

Suppose R denotes the rate of $\mathcal{R}_{\text{AERFC}}$, which *is the* radius of RF of AERFC and equals the rate of the 3×3 dilated filter $\mathcal{R}_{\text{Dilated}(3\times3)}$ plus 1. L_1, L_2, L_3, and L_4 denote four line segments, which restrict the range of p_1, p_2, p_3, and p_4, respectively. Specifically, L_1 can be described as

$$L_1 = \left\{ (x_1, y_1) | x_1 + y_1 = R, 0 \le x_1 \le \frac{R}{2}, \frac{R}{2} \le y_1 \le R \right\} \tag{2}$$

L_2 can be described as

$$L_2 = \left\{ (x_2, y_2) | x_2 + y_2 = R, \frac{R}{2} \le x \le R, 0 \le y \le \frac{R}{2} \right\} \tag{3}$$

L_3 is

$$L_3 = \left\{ (x_3, y_3) | x_3 \in [0, R], y_3 = \frac{R}{2} \right\} \tag{4}$$

and L_4 is

$$L_4 = \left\{ (x_4, y_4) | x_4 = \frac{R}{2}, y_4 \in [0, R] \right\}. \tag{5}$$

We use four nonnegative trainable variables V_1, V_2, V_3, and V_4 to control the locations p_1, p_2, p_3, and p_4, respectively. The location coordinates (x_1, y_1) of p_1 are defined as

$$\begin{cases} x_1 = \dfrac{R}{2} \times (1 - (V_1/(V_1 + 1))) \\ y_1 = \dfrac{R}{2} \times (1 + (V_1/(V_1 + 1))). \end{cases} \tag{6}$$

The location coordinates (x_2, y_2) of p_2 are described as

$$\begin{cases} x_2 = \dfrac{R}{2} \times (1 + (V_2/(V_2 + 1))) \\ y_2 = \dfrac{R}{2} \times (1 - (V_2/(V_2 + 1))). \end{cases} \tag{7}$$

The location coordinates (x_3, y_3) of p_3 are

$$\begin{cases} x_3 = \dfrac{R}{2} \\ y_3 = R \times (V_3/(V_3 + 1)). \end{cases} \tag{8}$$

The location coordinates (x_4, y_4) of p_4 are

$$\begin{cases} x_4 = R \times (V_4/(V_4 + 1)) \\ y_4 = \dfrac{R}{2}. \end{cases} \tag{9}$$

The location p_i can be updated with the trainable variables V_i changing.

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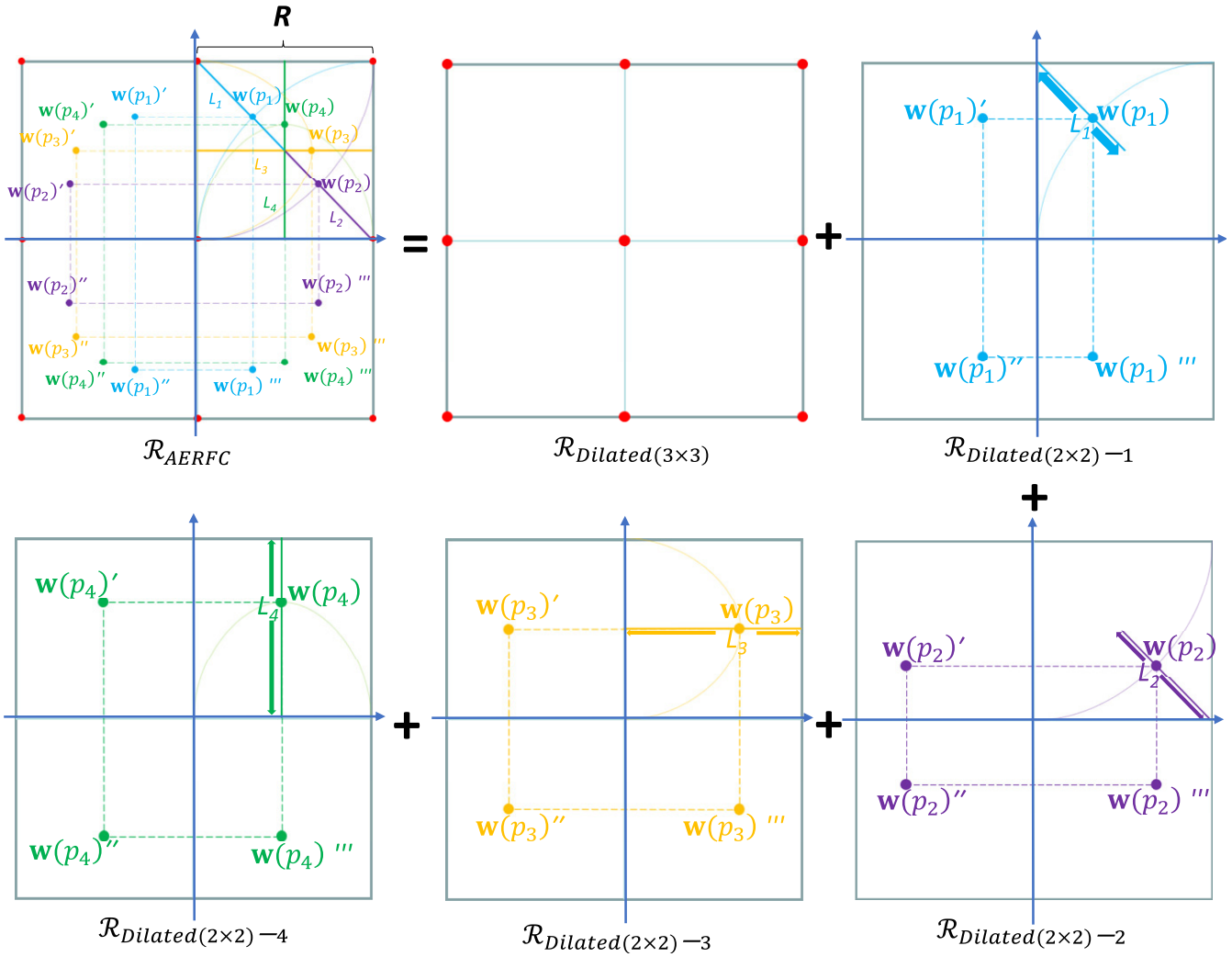


Fig. 1. AERFC filter $\mathcal{R}_{\text{AERFC}}$ is generated by filling 16 sampling weights $\mathbf{w}(p_i)$, $\mathbf{w}(p_i')$, $\mathbf{w}(p_i'')$, $\mathbf{w}(p_i''')$, where $i = 1, 2, 3, 4$, into a 3×3 dilated filter $\mathcal{R}_{\text{Dilated}(3\times3)}$. The location p_i is restricted on the corresponding line segment and controlled by the corresponding learnable nonnegative variable V_i. The locations p_i', p_i'', and p_i''' are generated by mirroring p_i. The resultant AERFC filter $\mathcal{R}_{\text{AERFC}}$ can be considered as a 3×3 dilated filter $\mathcal{R}_{\text{Dilated}(3\times3)}$ and four 2×2 dilated filters of variable size. $\mathcal{R}_{\text{Dilated}(2\times2)-i}$ $(i = 1, 2, 3, 4)$ represents a 2×2 dilated filter with sampling weights $\mathbf{w}(p_i)$, $\mathbf{w}(p_i')$, $\mathbf{w}(p_i'')$, $\mathbf{w}(p_i''')$.

In addition, the locations p_i', p_i'', and p_i''' are generated by mirroring p_i in the horizontal, vertical, and diagonal directions, respectively. For example, if the location coordinates of p_i are (x_i, y_i), the location coordinates of p_i', p_i'', p_i''' should be $(-x_i, y_i)$, $(-x_i, -y_i)$ and $(x_i, -y_i)$. The AERFC filter $\mathcal{R}_{\text{AERFC}}$ can be considered as the summation of the dilated filter $\mathcal{R}_{\text{Dilated}(3\times3)}$ and four 2×2 dilated filters $\mathcal{R}_{\text{Dilated}(2\times2)-i}$ $(i = 1, 2, 3, 4)$, whose size can be controlled by p_i, p_i', p_i'', and p_i'''.

The locations p_i', p_i'', and p_i''' are decided by p_i, and V_i controls the location of p_i. Therefore, V_i controls the location of p_i', p_i'', and p_i''', and determines the size and location of each $\mathcal{R}_{\text{Dilated}(2\times2)-i}$. In this way, V_i decides the effective RF of $\mathcal{R}_{\text{AERFC}}$.

Generally, the CNN updates the weights of each filter by calculating the partial derivative of the loss function to the weights in the filter. Variables must be included in the loss function to become learnable. To make V_i go into the loss function, we use V_i to divide each element in the corresponding feature map generated by convoluting the input feature map $\mathbf{P}$ with $\mathcal{R}_{\text{Dilated}(2\times2)-i}$ $(i = 1, 2, 3, 4)$ when implementing AERFC. The output map $\mathbf{M}$ of AERFC can be described by the following expression:

$$\mathbf{M} = \sum_{i=1,2,3,4} \frac{\mathcal{R}_{\text{Dilated}(2\times2)-i} * \mathbf{P}}{V_i} + \mathcal{R}_{\text{Dilated}(3\times3)} * \mathbf{P}. \quad (10)$$

In this equation, V_i becomes learnable and yields a better effective RF by adaptively inserting nonzero weights in appropriate locations.

For regular convolution, the contextual window size equals the size of the effective RF, which relies on filter sizes and strides of convolution and pooling operations. Even for the dilated filter, the size of the effective RF is fixed and remains the same after the neural network is determined [26]. The area outside the effective RF of a unit cannot affect the value of the unit [36].

Unlike the dilated filter, AERFC explicitly controls the sample locations of the convolution to compute feature responses. Thus, AERFCs dynamically capture rich contents and the high-level semantics of the region context conditioned on

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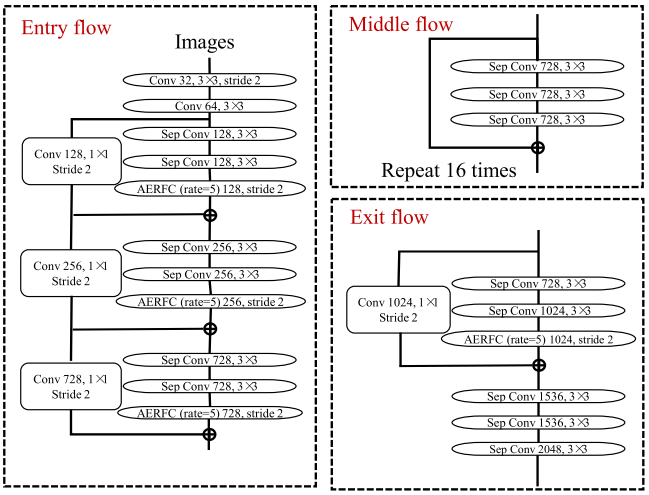


Fig. 2. Modified Xception with AERFCs.

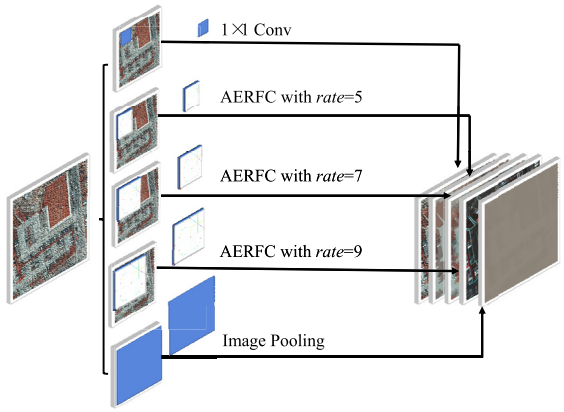


Fig. 3. Illustration of AERFC-SPP with three rates and five branches on VHR remote sensing images.

the input image, with dense small kernel sizes. It is more adaptive and flexible than traditional filters. Consequently, it can describe the internal variation of the input image.

B. Modified Xception With AERFC

We use a modified Xception with AERFC as the backbone of our network for better feature extraction. Xception [39], a well-known CNN, presents satisfactory results in image classification and object extraction via a fast calculation. The MSRA team [18] further modified Xception to form an aligned Xception, resulting in better object detection performance. Recently, Chen *et al.* [20] constructed an aligned modified Xception for semantic segmentation by modifying the aligned Xception with the dilated filter. For a more effective RF, we replace all the max pooling operations in Xception with AERFCs and use it as a network backbone for semantic segmentation. The details are listed as follows and are shown in Fig. 2: 1) we use the same entry flow and deeper layers as in [18] and [20]; 2) all max pooling operations in the Xception are replaced by "AERFC (rate = 5), stride 2" for a more effective RF, where the "AERFC (rate = 5), stride 2" denotes striding AERFC with rate = 2 and stride = 2; and 3) the batch normalization [40] and rectified linear unit (ReLU) activation are attached to each 3×3 depthwise separable convolution and each AERFC.

C. AERFC-SPP

Due to large-scale variations of the objects in VHR images, it is difficult to make accurate pixel-level predictions at a single scale. Multiscale representations are necessary for semantic segmentation with different RFs [37]. Therefore, AtrousSPP (ASPP) and pyramid scene parsing (PSP) networks [38] utilize predefined dilated filters with various dilated rates and pooling operations with various grids in SPP. Striding is used to capture long-range contextual information in the deeper blocks. As a result, the high-level features in the whole image can be summarized in the final small-resolution feature map. The size of the final feature map before the fully connected layers is 32 times smaller than the original image size. Unfortunately, small feature maps reduce detailed information. In addition, increasing the dilation rates not only expands the effective RF by covering large objects but also compromises the performance for small objects [19]. Moreover, ASPP and PSP cannot capture the internal scale variations of objects with arbitrary scales and sizes in VHR images [37].

Here, we construct AERFC-SPP to tackle these issues and extract more accuracy multiscale features for scene understanding. Especially, AERFCs are integrated with SPP for multiscale feature learning, replacing the dilated filters, as shown in Fig. 3. We stack AERFCs to capture multiscale information and generalize standard convolution. To maintain the spatial resolution of feature maps, the stride of the convolutional layer is set to 1 to avoid signal degeneration and to extract denser feature responses without many more extra parameters. AERFC is more adaptive and flexible with trainable location parameters. Hence, with the help of SPP, AERFC can describe more of the internal variation of the input image compared to the dilated filter. Furthermore, AERFC removes the gridding artifact by learning a few parameters in the dilated filter without significantly increasing the complexity.

D. AERFC ConvNet

CNN-based methods can be generally divided into two main categories: patch-based and pixel-to-pixel-based methods [41], [42]. The former predicts the class label of each pixel of the VHR images by classifying the region around that pixel. The classifier can be trained on small image patches and then used to classify all pixels with a sliding window. The procedure has high memory cost and low efficiency [42], [43]. The latter usually employs an FCN or encoder–decoder architectures, which replace the fully connected layer with interpolation, upsampling, or deconvolution. The latter can predict the class likelihoods of each pixel in a patch once and thus is more efficient. Our method employs the latter architecture.

The architecture of AERFC ConvNet is shown in Fig. 4. The AERFC ConvNet comprises a backbone CNN and AERFC in the SPP layer. The backbone CNN for feature extraction

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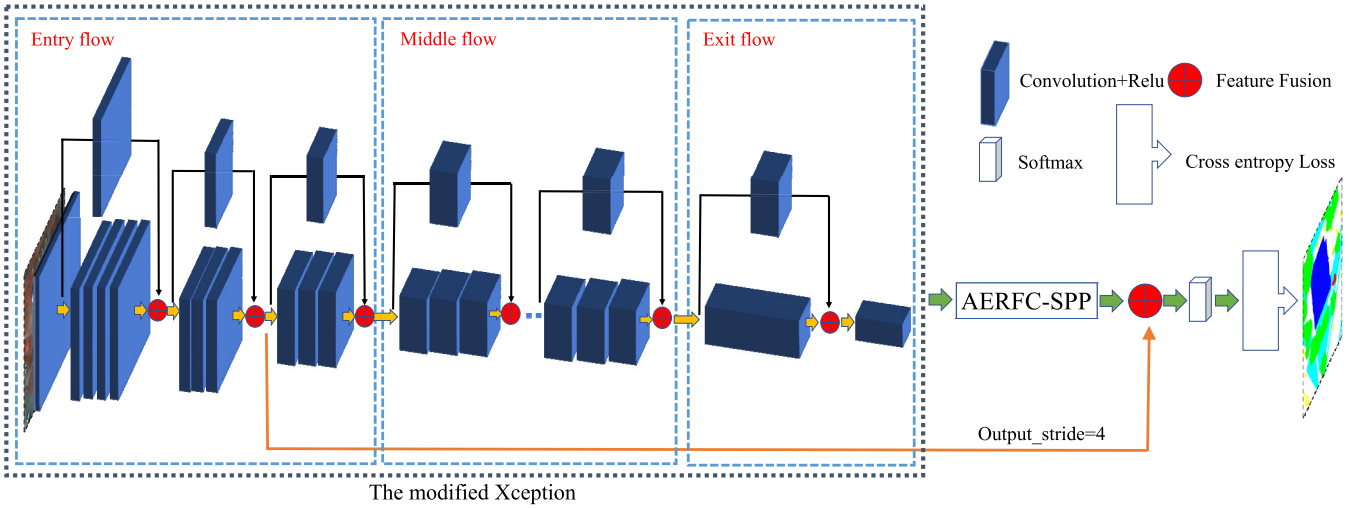


Fig. 4. Flowchart of the CNN with the proposed AERFC.

always employs ResNet, DenseNet, or Xception. Here, we use the modified Xception method. Each AERFC captures a specific scale feature representation, and different AERFCs obtain multiscale feature representations, which are more flexible and adaptive. AERFCs are arranged in parallel without losing the consistency of the feature representation over different scales. The multiscale feature representations are aggregated with the original feature extracted by Xception to compose the final feature representation for each pixel. These aggregated multiscale feature representations are fed into pixel-level predictors for semantic segmentation. AERFC is laid out in a cascade architecture, where different rates are adopted within different blocks.

IV. Experiments

The proposed AERFC provides a new way to improve the segmentation accuracy by introducing auxiliary into dilated convolution. To experimentally evaluate AERFC, two open remote sensing data sets from the International Society for Photogrammetry and Remote Sensing (ISPRS). To further verify the effectiveness of our method, two close-range photography images data sets are employed in the experiments, including Cityscapes and PASCAL VOC 2012. Next, we test the robustness of AERFC in terms of its parameter setting and structure. Then, the comparisons between AERFC and other related convolutions are shown. Finally, we discuss the effectiveness of AERFC.

A. Data Set Description

1) ISPRS Vaihingen Data Set [1]*:* The data set consists of a color infrared orthophoto and manually annotated ground truth labels. Thirty-three 3-band IRRG (infrared, red, and green) images are employed. Each of them has 2500*2000 pixels at a GSD of 9 cm. Sixteen tiles are used for training and validation, where five tiles (09 cm_area11, 09 cm_area15, 09 cm_area28, 09 cm_area30, 09 cm_area34) are used as the validation set in the experiments. The remaining 17 tiles are held for testing. In this work, DSM and NDSM data are not employed for training and inference. All results are obtained on the validation set, unless noted otherwise.

2) ISPRS Potsdam Data Set[2]*:* Thirty-eight 3-band IRRG images without DSM and NDSM data are employed in the experiments. Each image tile has a size of 6000×6000 pixels, with a GSD of 5 cm. Twenty-four tiles are annotated for training, where four tiles (image numbers 2_12, 4_11, 7_9, 7_11) are used for validation. The remaining 14 are held for testing. All statistics on Potsdam also rest on the validation set, unless noted otherwise.

3) PASCAL VOC 2012 Data Set [44]: It includes 20 object classes and one background class. The data set has 1464, 1449, and 1456 images for training, validation, and testing, respectively. The data set is augmented by the extra annotations provided by [45], resulting in 10 582 images for training. A total of 93 611 datum from MS-COCO data set [46] are converted into PASCAL VOC 2012 datum by keeping the same classes as PASCLA VOC 2012 and converting other classes into the background class. The original data set and the augmented data set are all used in our experiments.

4) Cityscapes Data Sets [47]: It includes fine annotation data sets and coarse annotation data sets. The fine annotation data set contains 5000 high-quality pixel-level finely annotated images from 50 European cities, of which 2975, 500, and 1525 images are used for training, validation, and testing, respectively. The other data set containing 20 000 images with coarse annotations can be used for training only.

B. Evaluation Metrics

For performance evaluation, three metrics are used, i.e., overall accuracy (OA), F1-score (F1), and mean intersection over union (mIoU). The OA is the total number of

[1]http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-vaihingen.html

[2]http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-potsdam.html

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TABLE I
SEGMENTATION RESULTS OF SEVERAL MODIFIED XCEPTION65S (MX-65) WITH DIFFERENT AERFC RATES ON THE VALIDATION SET OF ISPRS VAIHINGEN

Methods	F1 (%)					Mean F1 (%)	mIOU (%)
	Impervious surface	Building	Low vegetation	Tree	Car		
MX-65 with AERFCs of rate=3	88.7	93.0	80.9	85.1	67.6	83.1	71.1
MX-65 with AERFCs of rate=5	89.2	**93.5**	**81.5**	**85.6**	**67.9**	**83.5**	**71.9**
MX-65 with AERFCs of rate=7	**89.4**	93.1	81.2	85.2	67.1	83.2	70.8
MX-65 with AERFCs of rate=9	88.2	92.5	80.4	84.6	66.8	82.5	69.5

correctly classified pixels over the total number of all pixels as

$$\mathrm{OA} = \frac{tp + tn}{p + n} \tag{11}$$

where tp, tn, p, and n are the total number of true positives, true negatives, positives, and negatives, respectively.

F1 is the geometric mean between precision and recall for each class. The metric records the global accuracy when considering the over- and underpredictions for each class [48]. F1 over one class is defined as the harmonic mean of precision and recall

$$F1_i = 2 \times \frac{\mathrm{precision}_i \times \mathrm{recall}_i}{\mathrm{precision}_i + \mathrm{recall}_i} \tag{12}$$

$$\mathrm{precision}_i = \frac{tp_i}{tp_i + fp_i}, \quad \mathrm{recall}_i = \frac{tp_i}{tp_i + fn_i} \tag{13}$$

where tp_i, fp_i, and fn_i are the number of true positives, false positives, and false negatives attributed to class i, respectively.

Finally, the mIoU is the mean of the IoUs, which is the most popular evaluation metric for semantic segmentation. The IOU is defined as

$$\mathrm{IoU}(L_i \cup G_i) = \frac{|L_i \cap G_i|}{|L_i \cap G_i|} \tag{14}$$

where L_i is the set of ground truth pixels for the ith class, and G_i is the set of prediction pixels for the ith class. $\cup$ and $\cap$ denote intersection and union operations, respectively, and $||$ denotes the number of pixels in the set.

C. Data Augmentation

The hyperparameter number in CNNs is always large, while the number of training samples of the ISPRS Vaihingen and Potsdam data sets is relatively small. The relatively insufficient training samples will lead to overfitting. Overfitting can be overcome by data augmentation, including rescaling, slicing, and rotation. Data augmentation increases the diversity and quantity of training samples to improve the generalization ability of deep learning models [49].

The data augmentation steps of ISPRS data sets are listed as follows: 1) cropping raw images to generate a series of 769 $\times$ 769 patches with partial overlap; 2) rotating each patch with 90°, 180°, and 270°; 3) mirroring each patch along the y-axis; 4) executing the fuzzy operation for each patch; 5) adjusting the light for each patch; and 6) adding noise (Gaussian, salt, and pepper noise) to each patch. Table I lists detailed information on the experimental setting on the ISPRS Vaihingen and ISPRS Potsdam data sets. Fig. 5 shows several 769 $\times$ 769 patches from cropping the ISPRS Potsdam images. The crop size is 769 during both training and testing on the data set. During training, data augmentation is applied by randomly scaling the input patches (from 0.5 to 2.0) and random left-right flipping. During inference on the validation set, the flipped inputs are also added.

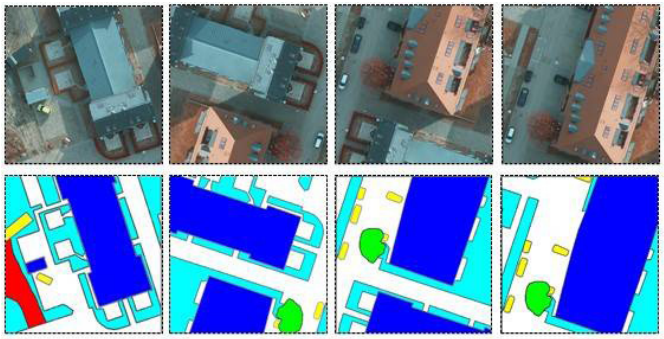


Fig. 5. Several 769 $\times$ 769 patches from cropping ISPRS Potsdam images. First line: original IRRG patches. Second line: ground truth maps.

D. Implementation Details

TensorFlow and Python 3.6 were used for all the experiments. All the experiments are conducted on a workstation equipped with two Intel i7-4960x CPUs, two 32 GB Rams, and 8 NVIDIA GTX Geforce 1080 Ti GPUs. The operating system is Ubuntu 18.04.

Due to the small size of the training sets of the ISPRS Vaihingen, ISPRS Potsdam, and PASCAL VOC 2012 data sets, the modified Xception65 (MX-65) is employed for them. The modified Xception71 (MX-71) is employed for the Cityscapes data set because of its relatively large size. This technique is initialized by the corresponding ImageNet pretrained Xception model from [50]. Specifically, a "poly" learning rate policy is also used; the momentum is set as 0.9, the weight decay is set as 0.00004, and the decay rate is set as 0.9997 for all experiments.

The initial learning rate is set as 0.001. For each AERFC, the four trainable nonnegative variables V_1, V_2, V_3, and V_4 are initialized by 1. The remaining weights of all networks are initialized with Xavier initialization, which is drawn from a Gaussian distribution with a zero mean and a specific variance of 0.01 [51]. The biases are initialized with ones. The convolutions are trained with the Adam optimizer to speed up training. For ISPRS Vaihingen and ISPRS Potsdam, the batch size is set to 8, and the cropped size is set to 769 $\times$ 769.

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TABLE II

SEGMENTATION RESULTS OF SEVERAL MODIFIED XCEPTION65 (MX-65) WITH DIFFERENT AERFC RATES ON THE VALIDATION SET OF ISPRS POTSDAM

Methods	F1 (%)					Mean F1 (%)	mIOU (%)
	Impervious surface	Building	Low vegetation	Tree	Car		
MX-65 with AERFCs of rate=3	**92.6**	95.1	86.5	82.6	72.6	85.9	72.5
MX-65 with AERFCs of rate=5	92.5	**95.6**	**86.9**	**83.1**	**72.9**	**86.2**	**73.0**
MX-65 with AERFCs of rate=7	90.2	93.1	85.4	81.4	69.3	83.9	70.4
MX-65 with AERFCs of rate=9	88.9	90.5	82.6	79.8	65.7	81.5	66.9

TABLE III

SEGMENTATION RESULTS OF SEVERAL MODIFIED XCEPTION65 (MX-65) WITH DIFFERENT AERFC RATES ON THE VALIDATION SET OF PSACAL VOC 2012. CLASSES 1–20 REPRESENT "AIRPLANE, BICYCLE, BIRD, BOAT, BOTTLE, BUS, CAR, CAT, CHAIR, COW, DINING TABLE, DOG, HORSE, MOTORBIKE, PERSON, POTTED PLANT, SHEEP, SOFA, TRAIN, AND TV MONITOR," RESPECTIVELY

Methods	IOU (%)																				mIOU (%)
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
MX-65 with AERFCs of rate=3	86.4	38.3	82.6	67.5	**75.1**	**90.7**	83.2	88.4	**49.1**	78.3	53.3	81.3	80.7	**82.1**	81.0	58.3	76.9	44.9	83.8	69.2	72.6
MX-65 with AERFCs of rate=5	**86.9**	**41.3**	**83.1**	**71.2**	74.8	90.6	**84.1**	88.5	48.9	**78.5**	54.1	80.9	**81.1**	81.7	**81.5**	58.6	**77.2**	**45.3**	84.1	**69.7**	**73.1**
MX-65 with AERFCs of rate=7	86.5	38.6	82.4	68.9	74.9	90.3	83.6	88.4	48.5	78.1	53.8	81.6	80.9	81.6	81.3	58.5	76.8	45.1	84.2	69.5	72.7
MX-65 with AERFCs of rate=9	86.6	40.3	83.0	68.3	74.5	90.4	83.7	**88.8**	48.2	78.0	**54.3**	80.6	80.6	81.9	81.2	**58.9**	77.1	45.2	**84.3**	69.5	72.8

TABLE IV

SEGMENTATION RESULTS OF SEVERAL MODIFIED XCEPTION71 (MX-71) WITH DIFFERENT AERFC RATES ON THE VALIDATION SET OF CITYSCAPES. CLASSES 1–19 REPRESENT "ROAD, SIDEWALK, BUILDING, WALL, FENCE, POLE, TRAFFIC LIGHT, TRAFFIC SIGN, VEGETATION, TERRAIN, SKY, PERSON, RIDER, CAR, TRUCK, BUS, TRAIN, MOTORCYCLE, AND BICYCLE," RESPECTIVELY

Methods	IOU (%)																			mIOU (%)
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	
MX-71 with AERFCs of rate=3	**97.6**	80.5	89.6	44.7	50.1	56.2	55.2	66.6	90.5	59.1	92.6	74.5	52.8	92.1	62.5	77.7	61.3	57.4	70.3	70.1
MX-71 with AERFCs of rate=5	97.5	**82.3**	**90.5**	45.1	**52.3**	**58.2**	**57.2**	**67.3**	90.4	**59.4**	**93.1**	**75.6**	**53.8**	**93.7**	**63.4**	**79.4**	**62.3**	**58.6**	**71.2**	**71.1**
MX-71 with AERFCs of rate=7	97.5	82.1	90.1	**47.6**	51.1	56.8	55.1	66.8	**90.6**	59.2	92.3	75.4	52.1	92.2	62.8	77.9	61.7	58.1	70.4	70.5
MX-71 with AERFCs of rate=9	97.4	81.3	90.2	45.6	50.5	57.1	55.4	66.9	89.8	58.7	92.3	74.9	52.9	92.3	62.7	78.1	61.1	57.3	70.2	70.2

We use parallel training with 8 GPUs, each of which addresses one patch. For PASCAL VOC 2012, the batch size is set to 24, and the cropped size is set to 513 $\times$ 513. Each GPU addresses three patches. For Cityscapes, the batch size is set to 8, and the cropped size is set to 513 $\times$ 1025. Each GPU addresses one patch.

E. Determination of AERFC Rates in the Backbone

We employ 80 K iterations for training, and batch normalization is used. For the last 40 K iterations, batch normalization parameters are frozen. AERCFCs with different rates are recorded in Tables I–IV. All AERCFCs in a neural network Xception have the same rate. From Tables I–IV, the best results come from the AERFCs of rate $=$ 5, and the worst results come from the AERFCs of rate $=$ 9. This result implies that the AERFCs of rate $=$ 5 achieve the best balance between the size of the whole RF and the amount of information captured in the RF of unit area, while AERFCs of rate $=$ 9 have the opposite balance.

F. Determination of Multi_Grid in AERFC-SPP

To extract context features effectively, it is very important to determine an appropriate contextual sampling distance in AERFC, which refers to the distance between the target position and the nearest neighbors. The maximum contextual sampling distance is determined by Multi_Grid.

After choosing MX-65 and MX-71 with AERFCs of rate $=$ 5 as the network backbone of the four data sets, the Multi_Grid in the three convolutional layers of AERFC-SPP is defined as $a \times (7; 5; 3)$, where $a = \{1, 2, 3, 4\}$. The networks with various Multi_Grid are trained with 100 k iterations, and batch normalization is used. For the last 50 k iterations, batch normalization parameters are frozen. During inference on the validation set, flipped inputs are added. The generated results are compared to determine the appropriate Multi_Grid, as shown in Tables V–VIII. From these tables, we can see that the best results come from the AERFC ConvNet equipped with AERFC-SPP of Multi_Grid $= \{14; 10; 6\}$.

G. Comparison With the Compared Methods on the Testing Sets of Four Data Sets

We choose MX-65 and MX-71 with AERFC rate $=$ 5, which are pretrained on ImageNet. Multi_Grid $= \{14; 10; 6\}$ for the four data sets, respectively. For testing on the ISPRS Vaihingen data set, the AERFC ConvNet is first trained with 100 K iterations on the train and validation (trainval) sets of the ISPRS Potsdam data set, and batch normalization is used. Then, AERFC ConvNet is trained with another 100 K

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TABLE V

SEGMENTATION RESULTS OF AERFC-SPPS WITH FOUR MULTI_GRID ON THE VALIDATION SET OF ISPRS VAIHINGEN

AERFC ConvNet with AERFC-SPP of different rates	F1 (%)					Mean F1 (%)	mIOU (%)
	Impervious surface	Building	Low vegetation	Tree	Car		
Rates:7, 5, 3	92.1	95.4	86.8	89.8	88.9	90.6	81.6
Rates:14, 10, 6	**93.7**	**96.2**	**87.3**	**90.6**	**89.5**	**91.5**	**83.5**
Rates:21, 15, 9	92.8	95.8	87.1	89.9	88.4	90.8	82.5
Rates:28, 20, 12	92.8	95.9	86.4	90.4	87.1	90.5	81.8

TABLE VI

SEGMENTATION RESULTS OF FOUR AERFC CONVNETS ON THE VALIDATION SET OF ISPRS VAIHINGEN

AERFC ConvNet with AERFC-SPP of different rates	F1 (%)					Mean F1 (%)	mIOU (%)
	Impervious surface	Building	Low vegetation	Tree	Car		
Rates:7, 5, 3	94.1	98.3	89.0	85.9	96.3	92.7	86.9
Rates:14, 10, 6	**94.4**	**98.6**	**89.2**	**86.7**	**97.1**	**93.2**	**87.5**
Rates:21, 15, 9	93.6	97.3	88.4	86.3	96.8	92.5	87.1
Rates:28, 20, 12	93.0	97.6	88.1	85.8	96.3	92.2	86.5

TABLE VII

SEGMENTATION RESULTS OF FOUR AERFC CONVNETS ON THE VALIDATION SET OF PSACAL VOC 2012. CLASSES 1–20 REPRESENT "AIRPLANE, BICYCLE, BIRD, BOAT, BOTTLE, BUS, CAR, CAT, CHAIR, COW, DINING TABLE, DOG, HORSE, MOTORBIKE, PERSON, POTTED PLANT, SHEEP, SOFA, TRAIN, AND TV MONITOR," RESPECTIVELY

AERFC ConvNet with AERFC-SPP of different rates	IOU (%)																				mIOU (%)
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
Rates:7, 5, 3	95.3	74.3	90.3	75.5	85.3	93.1	**88.3**	94.1	44.3	91.1	73.1	89.2	93.1	89.3	90.1	69.2	87.2	66.1	87.1	76.1	82.6
Rates:14, 10, 6	**95.5**	**74.6**	**91.2**	**75.8**	**86.1**	93.0	88.1	**95.1**	**45.6**	91.3	**73.9**	**90.1**	**94.2**	**89.8**	**90.5**	69.4	**88.3**	**67.2**	**87.2**	**77.2**	**83.2**
Rates:21, 15, 9	95.4	74.2	90.3	75.1	85.8	**93.2**	88.2	94.2	44.4	92.1	73.3	89.6	93.5	89.4	90.2	**69.6**	87.6	66.8	86.9	76.6	82.8
Rates:28, 20, 12	95.3	74.1	91.1	75.0	85.2	93.1	87.9	93.9	45.2	**91.7**	73.5	89.3	93.0	89.2	90.3	69.3	87.3	67.4	87.0	76.5	82.7

TABLE VIII

SEGMENTATION RESULTS OF FOUR AERFC CONVNETS ON THE VALIDATION SET OF CITYSCAPES. CLASSES 1–19 REPRESENT "ROAD, SIDEWALK, BUILDING, WALL, FENCE, POLE, TRAFFIC LIGHT, TRAFFIC SIGN, VEGETATION, TERRAIN, SKY, PERSON, RIDER, CAR, TRUCK, BUS, TRAIN, MOTORCYCLE, AND BICYCLE," RESPECTIVELY

AERFC ConvNet with AERFC-SPP of different rates	IOU (%)																			mIOU (%)
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	
Rates:7, 5, 3	96.5	84.6	91.1	60.1	61.3	63.1	71.2	76.1	90.4	70.2	**94.5**	**84.3**	70.6	94.0	77.6	89.2	80.1	68.2	75.3	78.9
Rates:14, 10, 6	96.1	**85.3**	**92.3**	**60.8**	**61.7**	**63.7**	**72.4**	**76.9**	**91.3**	**71.1**	93.9	83.8	**70.9**	**94.8**	**78.2**	**89.7**	**80.9**	**69.1**	**76.2**	**79.4**
Rates:21, 15, 9	**96.2**	85.1	91.2	60.3	61.2	63.3	71.1	76.2	90.5	70.3	94.1	83.9	70.4	94.3	77.6	89.1	80.5	68.5	75.6	78.9
Rates:28, 20, 12	96.0	84.3	91.5	60.5	61.5	63.1	70.9	75.8	90.1	70.6	93.2	83.4	70.3	94.1	77.2	89.3	79.8	68.5	75.4	78.7

TABLE IX

SEGMENTATION RESULTS ON THE TESTING SET OF THE ISPRS VAIHINGEN DATA SET

Methods	F1(%)					OA(%)
	Impervious surface	Building	Low vegetation	Tree	Car	
DST_2[43]	90.5	93.7	83.4	89.2	72.6	89.1
ONE_7[52]	91.0	94.5	84.4	89.9	77.8	89.8
BUAA_2[53]	91.6	94.7	84.0	89.3	86.0	89.9
BKHN11[54]	92.9	96.0	84.6	89.9	**88.6**	91.0
CASIA2[55]	93.2	96.0	84.7	89.9	86.7	91.1
NLPR3[56]	93.0	95.6	85.6	**90.3**	84.5	91.2
AERFC ConvNet	**93.1**	**95.9**	**85.9**	**90.3**	88.5	**92.0**

iterations on the ISPRS Vaihingen data set, where batch normalization parameters are frozen. For testing on ISPRS Potsdam, training is deployed in the reverse order.

For PASCAL VOC 2012, the AERFC ConvNet is pretrained with 100 K iterations on the MS-COCO variant data set and then with 50 K iterations on the augmented training set. During pretraining, batch normalization is used, and the initial learning rate is set as 0.01. Finally, the model is fine-tuned with 50 K iterations on the PASCAL VOC 2012 trainval set including 2913 images. During fine-tuning, batch normalization

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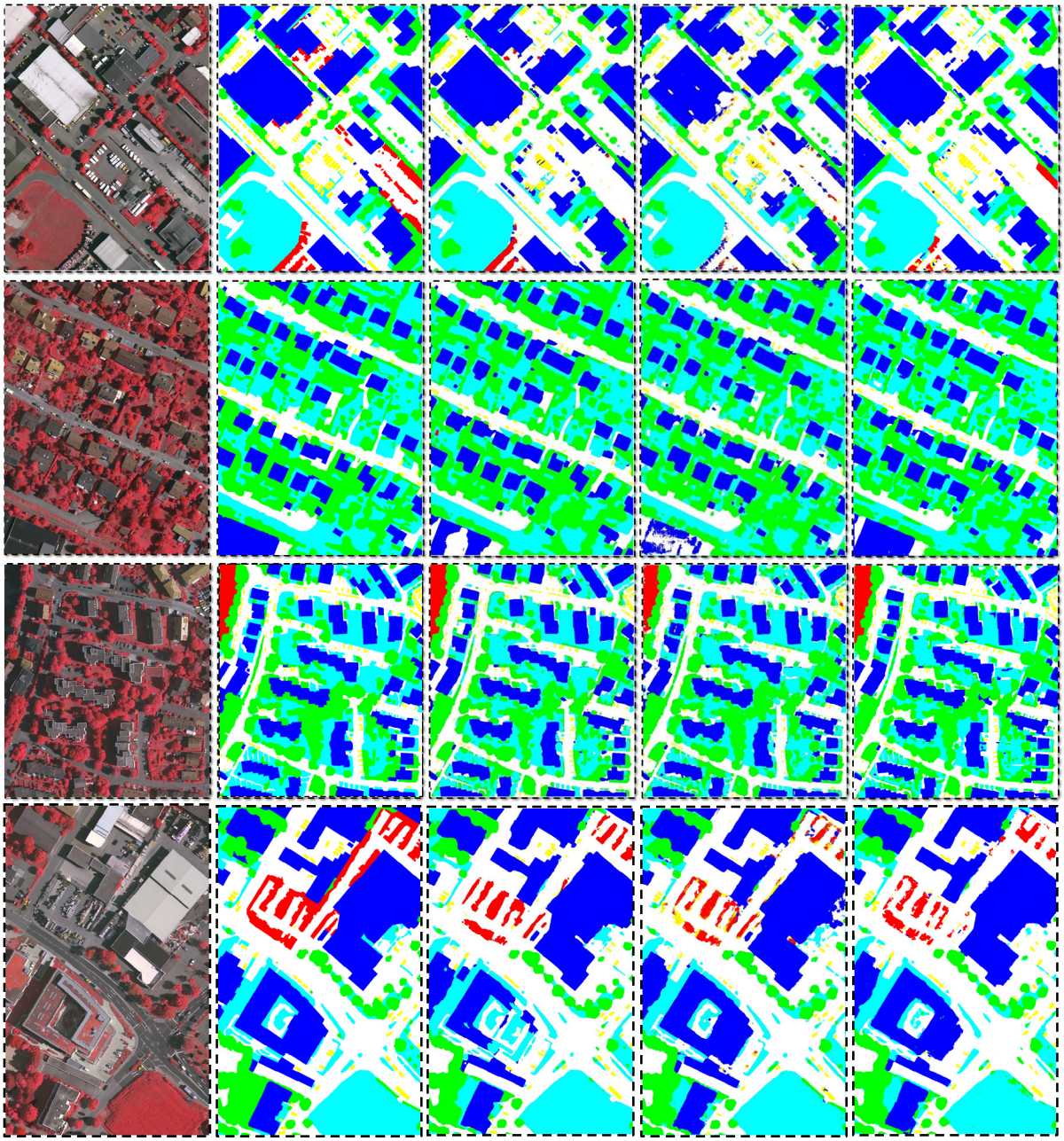


Fig. 6. Comparison of classification maps on image patches of the ISPRS Vaihingen data set. (a) Image patch. (b) Ground truth. (c) NLPR3. (d) HUSTW5. (e) Ours. There are six categories of objects in the prediction results. They are Impervious surface (white), Building (blue), Low vegetation (cyan), tree (green) and Clutter/Background (red).

parameters are frozen, and the base learning rate is set as 0.001.

For Cityscapes, we train AERFC ConvNet only on the trainval set of the fine annotation data set, and batch normalization is used. For the initial 100 K iterations, the initial learning rate is set to 0.01. For the next 100 K iterations, batch normalization parameters are frozen, and the base learning rate is set to 0.001.

1) Comparison With the Compared Methods on the ISPRS Vaihingen Testing Set: Fig. 6 and Table IX show qualitative and quantitative comparisons between various methods, respectively. The compared methods can be found on the official challenge website.[3] Unlike our method, the RFs of these compared methods cannot adaptively change to reach the optimization of performance. Furthermore, most of these methods used elevation data, namely, DMS and nDMS. However, as shown in Fig. 6, they are still less effective at segmenting confusing man-made objects, which have small spectral between-class variability and large within-class variability.

[3]http://www2.isprs.org/commissions/comm2/wg4/vaihingen-2d-semantic-labeling-contest.html

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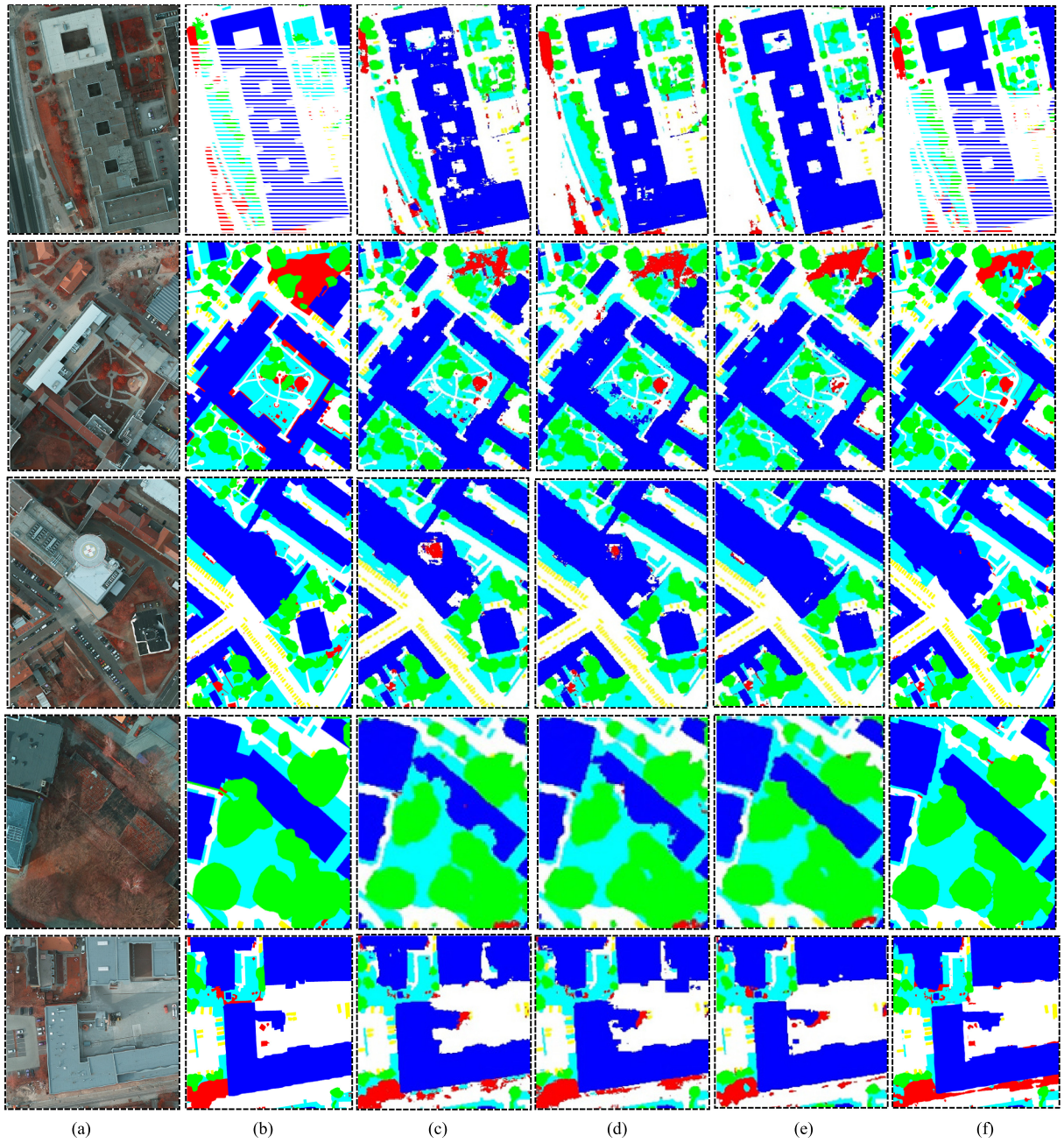


Fig. 7. Comparison of classification maps on image patches of the ISPRS Potsdam data set. (a) Image patch. (b) Ground truth. (c) NLPR3. (d) HUSTW5. (e) Ours. There are six categories of objects in the prediction results. They are Impervious surface (white), building (blue), low vegetation (cyan), tree (green) and Clutter/Background (red).

In contrast, our method yields much better results. From Table IX, it can be seen that our method achieves new state-of-the-art performance. The OA is 92.0%, which is 0.8% higher than the second-best OA. In addition, three out of six categories rank first in precision, including impervious surfaces, buildings, and low vegetation areas.

2) Comparison With the Compared Methods on the ISPRS Potsdam Testing Set: Fig. 7 and Table X show the qualitative and quantitative comparisons between different methods, respectively. These compared methods can be found on the official challenge website.[4] Most of them are listed in reference except two methods, HUSTW4 and SWJ_2, which may not be openly published. As introduced on the challenge website, HWSTW2 is implemented by combining DeconvNet [66] and U-Net [67], SWJ_2 is an adaptive approach for fusing multiscale features from ResNet101. It is also worth noting that DST_2 or _6 is the application of RefineNet in remote sensing. Unlike our method, the RFs of these compared methods

[4]http://www2.isprs.org/commissions/comm2/wg4/potsdam-2d-semantic-labeling.html

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TABLE X
SEGMENTATION RESULTS ON THE TESTING SET OF ISPRS POTSDAM DATA SET

Methods	F1(%)					OA(%)
	Impervious surface	Building	Low vegetation	Tree	Car	
RIT_2[57]	92.0	96.3	85.5	86.5	94.5	89.4
DST_2[43]	91.8	95.9	86.3	87.7	89.2	89.7
DST_6[43]	92.4	96.4	86.8	87.7	93.4	90.2
AZ3[58]	93.1	96.3	87.2	88.6	96.0	90.7
CASIA2[55]	93.3	97.0	87.7	88.4	96.2	91.1
AMA_1[59]	93.4	96.8	87.7	**88.8**	96.0	91.2
HUSTW4	93.6	97.6	**88.5**	88.8	94.6	91.6
SWJ_2	94.4	97.4	87.8	87.6	94.7	91.7
AERFC ConvNet	**94.4**	**98.0**	88.4	88.7	**96.3**	**92.1**

TABLE XI
SEGMENTATION RESULTS ON THE TESTING SET OF PSACAL VOC 2012

Method	IOU (%)																				mIOU(%)
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
DeepLabv3 [31]	96.4	76.6	92.7	77.8	87.6	96.7	90.2	95.4	47.5	93.4	76.3	91.4	97.2	91.0	92.1	71.3	90.9	68.9	90.8	79.3	85.7
EncNet [60]	95.3	76.9	94.2	**80.2**	85.2	96.5	90.8	96.3	47.9	93.9	80.0	92.4	96.6	90.5	91.5	70.8	93.6	66.5	87.7	80.8	85.9
DIS [23]	94.4	73.2	93.4	79.5	84.5	95.3	89.4	93.4	54.1	94.6	79.1	93.1	95.4	91.6	89.2	77.6	93.5	**79.2**	**93.9**	80.7	86.8
CFNet [61]	96.7	**79.7**	94.3	78.4	83.0	97.7	91.6	**96.7**	50.1	95.3	79.6	**93.6**	97.2	**94.2**	91.7	**78.4**	**95.4**	69.6	90.0	81.4	87.2
DeepLabv3+[20]	**97.0**	77.1	**97.1**	79.3	**89.3**	97.4	**93.2**	96.6	56.9	95.0	79.2	93.1	97.0	94.0	92.8	71.3	92.9	72.4	91.0	**84.9**	87.8
Ours	96.9	79.7	96.8	**80.2**	87.9	98.0	92.3	96.0	**58.2**	**95.8**	**83.0**	92.2	**97.3**	93.6	**93.0**	71.4	92.3	76.0	90.7	83.8	**88.2**

TABLE XII
SEGMENTATION RESULTS ON THE TEST SET OF CITYSCAPES

Method	IOU (%)																			mIOU (%)
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	
DenseASPP [62]	**98.7**	**87.1**	93.4	60.7	62.7	65.6	74.6	78.5	93.6	72.5	95.4	86.2	71.9	96.0	78.0	90.3	80.7	69.7	76.8	80.6
RestNet38 [63]	**98.7**	86.9	93.3	60.4	62.9	67.6	75.0	78.7	93.7	**73.7**	95.5	86.8	71.1	96.1	75.2	87.6	81.9	69.8	76.7	80.6
ACFNet [64]	**98.7**	**87.1**	**93.9**	60.2	**63.9**	71.1	**78.6**	81.5	**94.0**	72.9	**95.9**	**88.1**	**74.1**	**96.5**	76.6	89.3	81.5	72.10	**79.2**	81.8
InplaceABC [65]	98.4	85.0	93.6	**61.7**	**63.9**	67.7	77.4	80.8	93.7	71.9	95.6	86.7	72.8	95.7	79.9	93.1	89.7	72.6	78.2	82.0
DeepLabV3+	**98.7**	87.0	**93.9**	59.5	63.7	**71.4**	78.2	**82.2**	**94.0**	73.0	95.8	88.0	73.0	96.4	78.0	90.9	83.9	**73.8**	78.9	82.1
Ours	98.6	86.9	93.6	58.4	63.0	69.6	76.4	80.8	93.8	73.0	95.4	87.3	73.5	96.4	**81.5**	**93.5**	**90.4**	72.4	77.9	**82.23**

also cannot adaptively change to reach the optimization of performance. As shown in Fig. 7, our method can obtain more accurate segmentation objects, which have smoother boundaries and fewer defects. From Table X, it can be seen that our method achieves new state-of-the-art performance. The OA is 92.1%, which is 0.4% higher than that of the second-best OA. In addition, the precision of three out of six categories rank first with our method, including impervious surfaces, buildings, and cars.

3) Comparison With the Compared Methods on the PASCAL VOC 2012 Testing Set: Table XI shows the comparison results. From the table, it can be seen that our method also achieves state-of-the-art performance. The average accuracy (mIOU) is 88.2%, which outperforms DeeplabV3+ 0.4%. Six categories achieve leading segmentation accuracy. The results can also be found on the challenge official anonymous websites.[5]

4) Comparison With the Compared Methods on the Cityscapes Testing Set: Table XII shows the comparison results. From the table, it can be seen that our method also achieves state-of-the-art performance. The average accuracy (mIOU) is 82.23%, which is 0.13% better than the second. In addition, our method produces three first-rank categories in precision. The results from our method can also be found on the challenge official anonymous website.[6]

H. Size and Computational Efficiency of Our AERFC ConvNet

AERFC is implemented by depthwise separable convolution. Therefore, the size and computational time of AERFC ConvNet are greatly reduced. Table XIII compares the complexity of AERFC ConvNet with the state-of-the-art models.

[5]http://host.robots.ox.ac.uk:8080/anonymous/2NMN4E.html

[6]https://www.cityscapes-dataset.com/anonymous-results/?id=79f0d4ac2edd2ac32286e0b36d988dda30e156434099d877d20984fff6aef175

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TABLE XIII
COMPLEXITY COMPARISON WITH STATE-OF-THE-ART DEEP MODELS. "AX" REPRESENTS THE ALIGNED MODIFIED XCEPTION

	DenseASPP (DenseNet-161)	ACFNet (ResNet-101)	CASIA2 (ResNet-101)	DeepLabv3 (ResNet-101)	DeeplabV3+ (AX-65)	DeeplabV3+ (AX-71)	AERFC ConvNet (MX-65)	AERFC ConvNet (MX-71)
Model size	357 M	507 M	481 M	503 M	321 M	334 M	335 M	348 M
Runtime (sec)	1.5 sec/step	2.3 sec/step	1.8 sec/step	2.1 sec/step	0.95sec/step	1.1 sec/step	1.05sec/step	1.2 sec/step

The runtime (in seconds) spent on one training step (gradient update) is used to measure computational efficiency. The table also shows that AERFC ConvNet has competitive performance on both space and time complexity with respect to DeepLab V3+.

V. CONCLUSION

In this article, we propose AERFC to enlarge the effective RF for semantic segmentation of VHR remote sensing images without causing grid artifacts and greatly increasing additional parameters. AERFC adaptively controls the sampling location of convolution and automatically adjusts the effective RF, where the few parameters can determine the sampling locations in the nonzero filter taps of dilated filters. AERFC is also integrated with SPP to aggregate diverse multiscale features for exploring contextual information. The experimental results on four benchmark data sets confirm that AERFC outperforms the state-of-the-art methods in the number of quality metrics.

In future, AERFC would be applied to other computer vision tasks, such as object detection and instance segmentation, thanks to the adaptive effective RF. Experimental results on PASCAL VOC 2012 and Cityscapes in Tables XI and XII show that our method can achieve competitive performance on multiply classes, including dog, horse, sheep, cow, train, and bus. Even if the appearances of sheep, cow, and horse are similar to each other, our method can recognize them with high IOU. The results confirm the capability of our method on similar computer vision tasks.

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Xi Chen (Member, IEEE) received the B.Eng. degree from Chongqing University, Chongqing, China, in 2002, the M.Eng. degree from Shantou University, Shantou, China, in 2006, and the Ph.D. degree from Shanghai Jiao Tong University, Shanghai, China, in 2011.

He was with the Department of Information Engineering, School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin, China, as a Lecturer. He was also a Post-Doctoral Researcher with Purdue University, West Lafayette, IN, USA. He is a Researcher with the Key Laboratory of Geographic Information Science (Ministry of Education of China) and the School of Geographical Sciences, East China Normal University, Shanghai. His research interests include pattern recognition and image processing, as well as their applications to remote sensing images.

Dr. Chen serves as a Peer Reviewer for several international journals, such as the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS.

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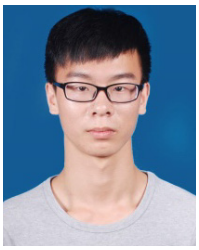
Zhiqiang Li received the Ph.D. degree in pattern recognition and intelligent systems from Shanghai Jiao Tong University, Shanghai, China, in 2009.

He is a Post-Doctoral Researcher with the Key Laboratory of Geographic Information Science (Ministry of Education of China) and the School of Geographical Sciences, East China Normal University, Shanghai. His research interests include pattern recognition and image processing, as well as their applications to remote sensing images.



Jie Jiang received the B.Eng. degree in software engineering from the China University of Geosciences, Wuhan, China, in 2019. He is pursuing the M.Eng. degree with East China Normal University, Shanghai, China.

His research interests include pattern recognition and image processing and their applications to remote sensing images.



Zhen Han received the B.Eng. degree from the North China University of Water Resources and Electric Power, Zhengzhou, Henan, China, in 2019. He is pursuing the M.Eng. degree with East China Normal University, Shanghai, China.

His research interests include scene recognition and time series prediction based on deep learning.



Shiyi Deng is pursuing the B.Eng. degree in electronic and information engineering with Hainan University, Hainan, China.

Her research interests include pattern recognition and image processing and their applications to remote sensing images.



Zhihong Li received the Ph.D. degree in human geography from the Institute of Geography Science, East China Normal University, Shanghai, China, in 1995.

He is an Associate Professor with the School of Geographic Sciences, East China Normal University. His major research interests include spatial data sharing and interoperability, WebGIS, and agricultural GIS.



Tao Fang received the B.S. and M.S. degrees in geology and survey from the Xi'an University of Science and Technology, Xi'an, China, in 1988 and 1991, respectively, and the Ph.D. degree in remote sensing and geographical information system from China University of Mining & Technology, Xuzhou, Jiangsu, in 1996.

He was employed as a Post-Doctoral Research Fellow in remote sensing and geographical information system with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, Wuhan, China, from 1996 to 1998. He was an Assistant Professor with the Department of Electronical Engineering and was employed as a Post-Doctoral Research Fellow in electronics with the HDTV Laboratory, University of Electronic Science and Technology of China, Chengdu, China, from 1999 to 2000, respectively. He is a Professor with the Institute of Image Processing and Pattern Recognition, Shanghai Jiao Tong University, Shanghai. His research interests include computer vision, statistical and structural pattern recognition, machine learning and data mining with applications to remote sensing classification, object recognition, change detection, land cover/land use, and so on.



Hong Huo (Member, IEEE) received the B.S. and M.S. degrees in computer application from the Jilin University of Technology, Jilin, China, in 1995 and 1998, respectively, and the Ph.D. degree in pattern recognition and intelligent systems from Shanghai Jiao Tong University, Shanghai, China.

Since 2000, she has been an Associate Professor with the Institute of Image Processing and Pattern Recognition, Shanghai Jiao Tong University. Her research interests include remote sensing and image classification, machine learning, and data mining with applications to remote sensing.



Qingli Li (Senior Member, IEEE) received the B.S. and M.S. degrees in computer science and engineering from Shandong University, Jinan, China, in 2000 and 2003, respectively, and the Ph.D. degree in pattern recognition and intelligent system from Shanghai Jiao Tong University, Shanghai, China, in 2006.

He is a Professor with the Shanghai Key Laboratory of Multidimensional Information Processing, East China Normal University, Shanghai, where he is researching medical imaging, pattern recognition, and image processing.



Min Liu received the Ph.D. degree in sedimentology from the Institute of Geology, Chinese Academy of Sciences, Beijing, in 1995.

He a Professor with the School of Geographic Sciences, East China Normal University, Shanghai, China. His major research interests include urban physical geography, natural hazards, and estuary biogeochemical cycles.

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